

Unique extension without a characteristic sequence: a sharp full-matrix dimension threshold

Candidate full counterexample to Question 1 of arXiv:1709.01649

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Status. Candidate full negative answer, likely valid. The question fails for an explicit pure state and a generating codimension-one operator system in $M_3(\mathbb{C})$. More generally, it fails in every M_d with $d \geq 3$, whereas it holds for every operator system in M_1 and M_2 . Thus dimension three is the sharp threshold among full matrix algebras.

1 The source question

Let A be a unital C^* -algebra and $S \subset A$ an operator system. A sequence (Δ_n) in A is *characteristic* for a state ψ if

- (a) $\|\Delta_n\| = 1$ for every n ;
- (b) $\psi(\Delta_n) \rightarrow 1$; and
- (c) $\limsup_{n \rightarrow \infty} \|\Delta_n^* a \Delta_n\| \leq |\psi(a)|$ for every $a \in A$.

Clouâtre asks in Question 1 of the section “Characteristic sequences”:

If ψ is a pure state on A with the unique extension property with respect to S , must there always exist a characteristic sequence in S for ψ ?

The original question and its surrounding discussion appear on PDF page 18 of the source and are reproduced in Figure 1.

Here the unique extension property (UEP) for a state means that it is the only state on A agreeing with its restriction to S . Every pure state of $M_d(\mathbb{C})$ has the form

$$\psi(a) = \langle a\xi, \xi \rangle = \text{Tr}(pa), \quad p = \xi\xi^*, \quad \|\xi\| = 1,$$

where p is its rank-one support projection.

2 Proof intuition

In finite dimension, the asymptotic object in the question cannot hide at infinity. A characteristic sequence has a norm-convergent subsequence. Its limit Δ must be killed on the left by $1 - p$, because condition (c) applied to $a = 1 - p$ has right-hand side zero. Hence Δ has range in the one-dimensional range of p . The norm and state normalization then force $\Delta = p$. Consequently, a characteristic sequence inside S exists if and only if $p \in S$.

so that $\varphi = \psi$ by part (2). We conclude that ψ has the unique extension property with respect to $\mathfrak{S}$. $\square$

The previous result shows that if a state ψ on $\mathfrak{A}$ admits a characteristic sequence in $\mathfrak{S}$, then ψ shares several properties with $\mathfrak{S}$ -peaking states. Indeed, (3) and (4) correspond to Lemma 3.1, while (2) says that ψ can be considered to be “asymptotically $\mathfrak{S}$ -peaking” on the sequence $(\Delta_n)_n$. Note however that typically a characteristic sequence does not consist of self-adjoint elements (see Example 4.6 below). We mention an outstanding problem that we have not been able to settle.

Question 1. Let $\mathfrak{A}$ be a unital C^* -algebra and let $\mathfrak{S} \subset \mathfrak{A}$ be an operator system. Let ψ be a pure state on $\mathfrak{A}$ that has the unique extension property with respect to $\mathfrak{S}$. Does there always exist a characteristic sequence in $\mathfrak{S}$ for ψ ?

A positive answer (even in the commutative case) would constitute a nice analogue of the classical fact that for uniform algebras, the Choquet boundary coincides with the set of peak points [30, Theorem II.11.3]. As mentioned before, this is known to fail for general function systems [43, page 43]. It is unclear to us at the moment whether the asymptotic peaking behaviour of characteristic sequences is flexible enough to remedy this defect.

We close this section by answering the previous question in the affirmative in some special cases. For that purpose, recall that an operator $T \in B(\mathcal{H})$ is said to be a *pure contraction* if $\|T\| \leq 1$ and

$$\lim_{n \rightarrow \infty} \|T^{*n}\xi\| = 0, \quad \xi \in \mathcal{H}.$$

Figure 1: Question 1 in arXiv:1709.01649v2, PDF page 18.

The problem is therefore to make p uniquely determined among density matrices by its values on S , without putting p itself in S . In M_3 this is possible with a trace-orthogonal hyperplane. Let

$$H = \text{diag}(1, 1, -2), \quad S = \{a \in M_3 : \text{Tr}(Ha) = 0\}, \quad p = E_{11}.$$

Every density matrix agreeing with p on S lies on the affine line $p + tH$. Positivity of its second and third diagonal entries requires both $t \geq 0$ and $t \leq 0$, so $t = 0$. Thus UEP holds, but $p \notin S$.

In M_2 there is no room for this two-sided positivity trap. If $p \notin S$, a trace-orthogonal self-adjoint direction through p always points into the state space for one sufficiently small choice of sign. It produces a second state with the same restriction, contradicting UEP. This supplies the sharp dimension threshold.

3 The support-projection criterion

Lemma 1. Let $S \subset M_d(\mathbb{C})$ be an operator system, let $\psi(a) = \langle a\xi, \xi \rangle$ be a pure state, and let $p = \xi\xi^*$. Then ψ admits a characteristic sequence contained in S if and only if $p \in S$.

Proof. If $p \in S$, take $\Delta_n = p$ for all n . Then $\|p\| = 1$, $\psi(p) = 1$, and

$$pap = \psi(a)p, \quad \|pap\| = |\psi(a)|,$$

so all three defining conditions hold.

Conversely, suppose $(\Delta_n) \subset S$ is characteristic. The unit ball of the finite-dimensional space S is compact, so some subsequence converges in norm to an element $\Delta \in S$. Conditions (a) and (b) give $\|\Delta\| = 1$ and $\psi(\Delta) = 1$. Put $q = 1 - p$. Since $\psi(q) = 0$, condition (c), followed along the convergent subsequence, yields

$$\|\Delta^*q\Delta\| = 0.$$

Thus $q\Delta = 0$, or equivalently $p\Delta = \Delta$. Moreover $p\Delta p = \psi(\Delta)p = p$, and hence $\Delta p = p$. Put $x = \Delta q$. Then $\Delta = p + x$, the initial spaces of p and x are orthogonal, and the range of x lies in the one-dimensional range of p . Consequently

$$\Delta\Delta^* = p + xx^* = (1 + \|x\|^2)p.$$

Since $\|\Delta\| = 1$, this forces $x = 0$. Hence $\Delta = p \in S$. $\square$

4 The counterexample and the sharp threshold

Theorem 2. *The source question has a negative answer already for $A = M_3(\mathbb{C})$. More precisely:*

- (i) *for every $d \geq 3$, there is a generating operator system $S \subset M_d(\mathbb{C})$ and a pure state ψ having the UEP with respect to S but no characteristic sequence in S ;*
- (ii) *for $d = 1, 2$, every pure state on $M_d(\mathbb{C})$ having the UEP with respect to an operator system S does admit a characteristic sequence in S .*

Thus $d = 3$ is the least full-matrix dimension in which the implication fails.

Proof. Fix $d \geq 3$ and set

$$p = E_{11}, \quad H = \text{diag}(1, 1, -2, 0, \dots, 0), \quad S = \{a \in M_d(\mathbb{C}) : \text{Tr}(Ha) = 0\}.$$

Because $H = H^*$ and $\text{Tr } H = 0$, the complex hyperplane S is self-adjoint and contains the identity; hence it is an operator system. It contains every off-diagonal matrix unit E_{ij} . Products $E_{ij}E_{ji} = E_{ii}$ then show that $C^*(S) = M_d(\mathbb{C})$. Also

$$\text{Tr}(Hp) = 1,$$

so $p \notin S$.

Let $\psi(a) = \text{Tr}(pa) = a_{11}$. This is a pure state. Suppose a state $\varphi(a) = \text{Tr}(\rho a)$ agrees with ψ on S , where $\rho \geq 0$ and $\text{Tr } \rho = 1$. The trace pairing on $M_d(\mathbb{C})$ is nondegenerate, and S is the kernel of the nonzero functional $a \mapsto \text{Tr}(Ha)$. Therefore its annihilator is $\mathbb{C}H$, and agreement on S gives

$$\rho - p = tH$$

for some scalar t . Self-adjointness makes t real. The second and third diagonal entries of $\rho = p + tH$ are respectively t and $-2t$. Positivity forces both to be nonnegative, hence $t = 0$. Thus $\rho = p$, proving that ψ has the UEP. Since $p \notin S$, Lemma 1 proves that ψ has no characteristic sequence in S . This proves (i), including the explicit M_3 counterexample.

For (ii), the $d = 1$ case is immediate. Let $d = 2$, let p be the rank-one support of a pure state ψ , and suppose that $p \notin S$. Regard $M_2(\mathbb{C})_{\text{sa}}$ as a real inner-product space under $(x, y) \mapsto \text{Tr}(xy)$. Since $p \notin S_{\text{sa}}$, finite-dimensional separation gives a self-adjoint H such that

$$\text{Tr}(Hs) = 0 \quad (s \in S), \quad a := \text{Tr}(Hp) \neq 0.$$

The first equality holds for all complex $s \in S$ by decomposing s into self-adjoint real and imaginary parts. Since $1 \in S$, we also have $\text{Tr } H = 0$. In a basis with $p = E_{11}$, write

$$H = \begin{pmatrix} a & z \\ \bar{z} & -a \end{pmatrix}.$$

Choose a nonzero real t of sign opposite to a and with magnitude so small that

$$|t| < \min \left\{ \frac{1}{|a|}, \frac{|a|}{a^2 + |z|^2} \right\}.$$

Then $\rho = p + tH$ has trace one, both diagonal entries are positive, and

$$\det \rho = -ta - t^2(a^2 + |z|^2) > 0.$$

Hence ρ is a positive definite density matrix distinct from p . Moreover $\text{Tr}((\rho - p)s) = t \text{Tr}(Hs) = 0$ for all $s \in S$, so ρ defines a second state agreeing with ψ on S . Thus $p \notin S$ contradicts the UEP. We conclude that UEP forces $p \in S$ in M_2 , and the constant sequence $\Delta_n = p$ is characteristic by Lemma 1. $\square$

5 Verification, scope, and novelty

The proof reduces every asymptotic statement to finite-dimensional norm compactness. The M_d UEP calculation uses all states, not merely pure states: every state is represented by a density matrix, and the annihilator of the displayed hyperplane is exactly one-dimensional. The example's operator system generates the ambient matrix algebra, so the failure is not caused by placing S inside a proper C^* -subalgebra.

The theorem is a full counterexample to the unrestricted yes/no question and a complete classification by dimension for full matrix algebras. It does not classify arbitrary finite-dimensional direct sums or infinite-dimensional C^* -algebras, which is unnecessary for the negative answer.

A bounded novelty sweep searched the run's lightweight indexes and primary arXiv sources for the exact question, characteristic sequences, UEP, and pure-state restrictions. It found the source paper and Clouâtre's later paper on restrictions of pure states and excision [2], but no explicit answer to Question 1 or the matrix construction above. A different existing run packet associated with arXiv:2306.02365 concerns failure of a distinct pure-extension property and is not the present result. Novelty confidence is therefore moderate rather than an assertion of exhaustive priority. No external theorem is used in the proof.

Human review checklist. Check the limiting argument in Lemma 1; verify the one-dimensional trace annihilator and the two diagonal positivity constraints in the $d \geq 3$ example; and verify the determinant sign and small- t bounds in the M_2 converse. These are the only substantive proof points.

References

- [1] R. Clouâtre, *Non-commutative peaking phenomena and a local version of the hyperrigidity conjecture*, arXiv:1709.01649v2 (2018), especially Question 1, PDF p. 18.
- [2] R. Clouâtre, *Restrictions of pure states to subspaces of C^* -algebras*, arXiv:2306.02365 (2023; later revisions).